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Ruthenium brachytherapy for uveal melanomas: Factors affecting the development of radiation complications

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ABSTRACT

PURPOSE: To investigate how treatment complications are related to dosimetric parameters after ruthenium-106 brachytherapy for uveal melanoma, in a large, clinically homogeneous population.

METHODS AND MATERIALS: A retrospective review was performed to evaluate patients affected by small and medium size uveal melanoma, treated with ruthenium-106 brachytherapy from December 2006 to December 2014. We excluded tumors with posterior margin within 1 mm from the edge of the optic disc and foveola. Main outcome measures were occurrence and time to radiation-related maculopathy, cataract, and optic neuropathy. Secondary end points included local recurrence and distant metastases. Kaplan−Meier analysis with log-rank test and univariate Cox proportional hazards model at 3 years were performed to identify the covariates affecting the outcome of radiation complications.

RESULTS: Two hundred thirty-nine patients, with sufficient data to evaluate the end points, were enrolled. Three years after plaque treatment, radiation maculopathy was found in 61 (25.5%) patients, cataract developed in 10 patients (4.2%) receiving a dose of 27 Gy or higher to the lens, and optic neuropathy was observed in 13 patients (5.4%) with an optic nerve dose exceeding 50 Gy and tumor proximity to optic disc of less than 4 mm. Tumor recurrences and tumor-related metastasis were found respectively in 20 (8.36%) and 14 (5.85%) patients.

CONCLUSIONS: Radiation maculopathy occurs within a median time of 31 months in 25% of cases after plaque treatment for uveal melanoma. The most significant risk factors are total dose and distance of tumor margin from the fovea. If the distance is greater than 11.22 mm, no signs of retinal damage are detected. © 2017 American Brachytherapy Society. Published by Elsevier Inc. All rights reserved.

Keywords: Plaque brachytherapy; Interventional radiotherapy; Radiation maculopathy; Ruthenium^{106}; Uveal melanoma

Introduction

Brachytherapy (interventional radiotherapy) is the most commonly used form of radiotherapy for uveal melanoma, by use of isotopes of iodine-125 (I-125) and ruthenium-106 (Ru-106), it achieved a local control rate in the range of 88−98% at 5 years (1).

Radiation retinopathy represents the most common ocular side effect after brachytherapy for uveal melanoma; it has been reported to occur in 10−63% of postplaque eyes, with higher rates associated with larger tumors, increased radiation dosage, and posterior location (2).

Radiation retinopathy is characterized by a slowly progressive, occlusive vasculopathy with a delayed onset after irradiation and usually leads to irreversible visual impairment (3).

The area of the eye that appears to be most sensitive to radiation damage is the posterior pole.

Radiation maculopathy typically develops when radiation exposure extends beyond tissue tolerance. The earliest and

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most common clinical feature of radiation maculopathy is macular edema, which can lead to ischemic maculopathy (4).

Each radionuclide offers different energies and intraocular distributions; the goal of brachytherapy treatment is to deliver a curative dose to the tumor while offering the least possible radiation to normal ocular structures.

Ru-106 is a beta emitter, with a steep dose falloff and lower penetration depth compared with I-125; Ru-106 plaque delivers a much higher scleral to apex dose than I-125, while the dose to the contralateral structures is reduced.

It has been shown that for tumors not exceeding 5 mm in thickness, the use of Ru-106 plaques has the potential to reduce the radiation dose to nearby normal structures and possibly lower the risk of radiation-induced visual loss (5).

We reviewed our experience with Ru-106 plaque interventional radiotherapy to treat uveal melanoma with respect to risk factors that led to the development of radiation retinopathy and investigated the relationship between radiation complications and dosimetric parameters.

Methods and materials

Study design

The records of patients treated at the Ocular Oncology Unit of the Catholic University of Rome, for primary uveal melanoma, with Ru-106 plaque brachytherapy, between 2006 and 2014, were retrospectively reviewed. All data were obtained from the hospital's multidivisional electronic database—"Spider's Net." To ensure patient's privacy, clinical data were elaborated electronically through the Consortium for Brachytherapy Data Analysis Storage System (6, 7). The relationship between complications and dosimetric parameters was investigated.

Patients with posterior tumor margin within 1 mm from the edge of the optic disc and foveola were excluded because of the high probability of developing radiation sequelae. Four patients were excluded from analysis as the melanoma had been previously treated with other modalities.

The population studied consisted of 239 patients treated with the Ru-106 plaque.

The diagnosis was made on the basis of the ophthalmoscopic evaluation, A−B scan ultrasonography, and fundus photography.

A metastatic workup, including liver function test and liver ultrasound was performed, before treatment, for each patient and was negative in all cases.

This retrospective clinical research study was carried out with permission from the institutional review board of the Catholic University.

Interventional radiotherapy protocol

The baseline patient data including demographic features such as age and gender, systemic illness (diabetes mellitus, hypertension), prior history of malignancy, and prior ophthalmological conditions were recorded. Pretreatment tumor data included tumor shape and location, TNM classification of malignant tumors, and tumor thickness, basal diameter and tumor distance to the optic disc and fovea measured by ultrasound using A−B scan standardized technique.

Each case was evaluated in a weekly multidisciplinary meeting including an ocular oncologist, a radiation oncologist, and a clinical physicist. The radioisotope used was Ru-106, and the prescription dose was 100 Gy to the tumor apex (8).

The dose calculation was performed using the software Plaque Simulator (BEBIG Isotopen-und Medizintechnik GmbH, Berlin, Germany), calculating the estimated dose to organs at risk, according to the activity of the plaque used. The size of the episcleral plaque was selected to provide 1-mm margin in all directions around the tumor edge. The plaque was applied to the patient under local anesthesia, and the lesion localization was performed by the use of transillumination or indirect ophthalmoscopy, based on tumor location.

Intraoperative plaque placement confirmations were performed by experienced diagnostic ophthalmic echographers to ensure that the plaque was centered on the tumor base and all tumor margins were covered by the plaque.

Followup examinations were scheduled 15 days after the removal of the plaque, at 4-month intervals for up to 2 years and at 6 months intervals thereafter.

Statistical analysis and end points

Kaplan−Meier analysis with log-rank test and univariate Cox proportional hazards model at 3 years were performed to identify the covariates, which may have an effect on the onset and outcome of maculopathy. A p-value ≤ 0.05 was considered significant.

The time independence of predictors (Cox assumption) was verified using Schoenfeld residuals. The statistical analysis was performed using R version 3.3.1.

Primary end points included the incidence, the time of appearance and the factors related to the development of radiation damage, identified by ophthalmoscopy, optical coherence tomography, and fluorescein angiography. Secondary end points were both local, represented by tumor regrowth, and distant failure, defined as metastatic disease.

Results

Between December 2006 and December 2014, 239 patients (239 eyes) received Ru-106 brachytherapy treatment. The baseline patient demographics, clinical features, and tumor characteristics are summarized in Table 1.

The median largest basal tumor diameter was 9.86 mm (range 3.6−16.1 mm), and the median tumor thickness was 3.29 mm (range 2−5.9 mm).

Table 1
Baseline patient demographics, clinical features, and tumor characteristics

Characteristic	Value (%)
Sex	
Female	135 (56.4)
Male	104 (44.6)
Laterality	
Right	127 (53.1)
Left	112 (46.8)
Age	67 (median) (range 17–92)
Systemic disease	
Systemic diabetes	28 (11.7)
Systemic hypertension	111 (46.4)
Shape	
Dome	229 (95.8)
Mushroom	7 (2.9)
Plateau	2 (0.8)
Pigmentation	
Pigmented	219 (91.6)
Amelanotic	20 (8.4)
Location	
Iridic	1 (0.4)
Iridociliary	4 (1.6)
Iridociliochoroidal	5 (2.1)
Ciliochoroidal	5 (2.1)
Choroidal	224 (93.7)
Quadrant	
Superior	74 (30.9)
Temporal	75 (31.4)
Inferior	57 (23.8)
Nasal	32 (13.8)
Tumor parameters	
Tumor thickness	3.29 mm (median) (range 2–5.9)
Largest basal diameter	9.86 mm (median) (range 3.6–16.1)
Tumor distance to the edge of optic disc	10.62 mm (median) (range 1–22.53)
Tumor distance to the foveola	10.03 mm (median) (range 1.11–22)
Subretinal fluid	61 (25.5)
TNM classification of malignant tumors	
T1	113 (47.2)
T2	123 (51.46)
T3	3 (1.25)

The median distance of the posterior margin of the tumor to the edge of the optic disc was 10.62 mm (range 1–22.53 mm), and the median distance to the fovea was 10.03 mm (range 1.11–22 mm). Sixty-one (25.5%) patients had retinal detachment secondary to tumor or subretinal fluid. No patient had known extraocular extension before plaque placement.

According to the eighth edition of the American Joint Committee on Cancer staging system, 113 (47.3%) patients were T1, 123 (51.5%) T2, and 3 (1.25%) T3 at diagnosis.

Depending on tumor basal diameter, the following plaque manufactured by Bebig, Berlin, Germany, were used: CCA (plaque diameter [PD]:15.3 mm) in 84 (35%) patients; CCD (PD: 17.9 mm) in 49 (20.5%); CCB (PD: 20.2 mm) in 55 (23%); and COB (PD:19.8 mm) in 51 (21.3%) patients.

The radiation characteristics are detailed in Table 2.

Followup ranged from 2 to 107 months (median 48 months); all patients included in the study had regular followup.

Radiation complications

Radiation maculopathy was found in 61 (25.5%) patients after 3 years from plaque treatment. By using Kaplan–Meier survival curves, we found that the proportion of patients in whom radiation maculopathy was observed was 9.6% (95 percent confidence level: 5.7–13.3%) of patients at 1 year, 19.9% (95 percent confidence level: 14.3–25.2%) at 2 years, and 25.1% (95 percent confidence level: 18.7–31.0) at 3 years.

The median time to develop maculopathy was found to be 31 months after plaque treatment and the range was 0–104 months.

Results of the univariate Cox proportional hazards regression analysis showed that the factors predictive of the development of radiation maculopathy were tumor height ($p = 0.037$), dose to the fovea ($p < 0.001$), dose rate to the fovea ($p < 0.001$), distance to the fovea ($p < 0.001$), tumor location (quadrant) ($p < 0.05$), tumor volume ($p = 0.022$), and the presence of diabetes ($p = 0.011$) and subretinal fluid ($p = 0.001$)(Table 3).

We examined radiation maculopathy outcome in relation to fovea dose. As shown in Fig. 1, we identified a threshold dose to the fovea of 50 Gy; below this value, the risk associated to the development of radiation maculopathy was 12%, and above it, the associated risk was 46% on average (range 31–60%).

The incidence of maculopathy for patients that received a dose to the fovea lower than 50 Gy was significantly lower than other intervals of dose.

Examining the distance to the fovea as a possible risk factor, the Fisher exact test revealed that the risk of developing radiation maculopathy was significantly associated ($p < 0.001$) with the distance of the tumor margin from the fovea. In particular, we found a cutoff of 11.22 mm (distance between the posterior tumor margin and fovea); beyond this value, no patients developed radiation maculopathy (Fig. 2).

Table 2
Radiation parameters of ruthenium-106 plaque radiotherapy

Variable	Median	Range
Tumor apex dose Gy	99.99	84.99–133.8
Tumor apex dose rate cGy/h	186.1	56.99–389.5
Macula dose Gy	25.78	0–568.2
Macula dose rate cGy/h	28.95	0–263
Optic disc center dose Gy	16.8	0–261.1
Optic disc center dose rate cGy/h	28.95	0–263
Lens dose Gy	0.06	0–83.52
Lens dose rate cGy/h	0.12	0–280.8
Sclera dose Gy	268.4	160.8–709
Sclera dose rate cGy/h	520.2	157.0–858.8

Table 3
Statistically significant ($p < 0.05$) predictors for maculopathy occurrence in the univariate Cox proportional hazards regression analysis

Predictor	*p*-value
Tumor height	0.037
Dose to the fovea	<0.001
Dose rate to the fovea	<0.001
Distance to the fovea	<0.001
Tumor location (quadrant)	<0.05
Tumor volume	0.022
Diabetes	0.011
Subretinal fluid	0.001

In our study, only 10 (4.2%) of the phakic patients plaqued for anterior uveal melanoma that had received a dose of 27 Gy or higher to the lens developed a secondary cataract.

Of the 239 patients, 13 (5.4%) developed optic neuropathy, after a median of 41 months after plaque.

Patients with optic neuropathy had received a mean dose to the edge of the optic disc of 95.08 Gy (range 50.39−134.9), and in all cases, the posterior margin of the tumor was less than 4 mm from the disc margin.

Local tumor control

Twenty (8.4%) of 239 eyes developed recurrent tumor growth. The estimated local tumor recurrence rates at 12, 24, and 36 months after irradiation were 2.2%, 5.2%, and 8.3%, respectively.

Among the 20 cases of local recurrences, 13 were treated with enucleation, one with transcleral resection and three with proton beam; in three cases, a second plaque controlled the tumor without the need for enucleation.

Metastatic disease

Of the 239 patients, 14 (5.85%) developed distant metastatic disease. Kaplan−Meier estimates demonstrated that the metastasis-free survival was 98.7% at 12 months, 96.7% at 24 months, and 96.7% at 36 months. Kaplan−Meier estimates showed that the disease-specific survival was 97.8% at 12 months, 92.8% at 24 months, and 90% at 36 months.

Discussion

Maculopathy, cataract, and optic neuropathy are the most common treatment-related morbidities for patients with uveal melanoma; they can be seen months after treatment and result in progressive visual loss.

Wen *et al.* reported that, among different studies using I-125, maculopathy incidence varies between 10% and 63%, cataract varies between 8% and 83% by 5 years after plaque, and optic neuropathy has been reported in up to 16% of patients (9).

For patients treated with Ru-106 plaque, the incidence reported varies between 19.6% and 50% for radiation maculopathy, 2.5−37.2% for cataract, and 2.2−29.4% for optic neuropathy (10−14).

In our work, radiation maculopathy developed in 25.5% of patients, cataract was observed in 4.2% of patients, and optic neuropathy in 5.4% after 3 years from Ru-106 plaque.

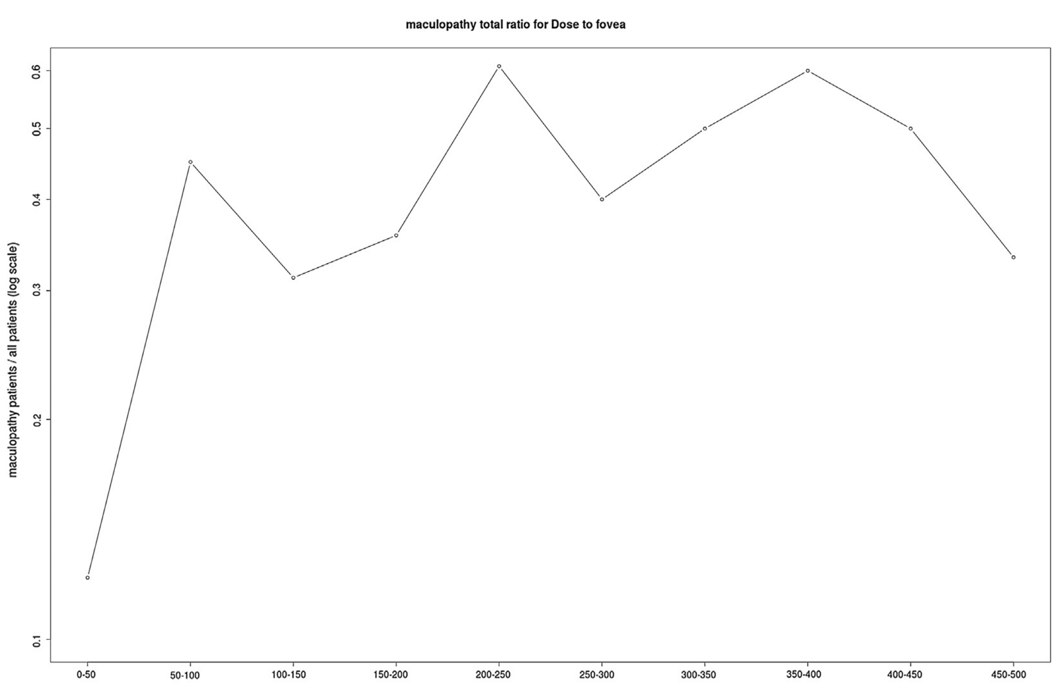

Fig. 1. Graph showing the incidence of radiation maculopathy as a function of dose to the fovea. Y axis: number of patients affected by maculopathy as a fraction of total number of patients (log scale); X axis: dose (Gray) delivered to the fovea (50-Gray intervals).

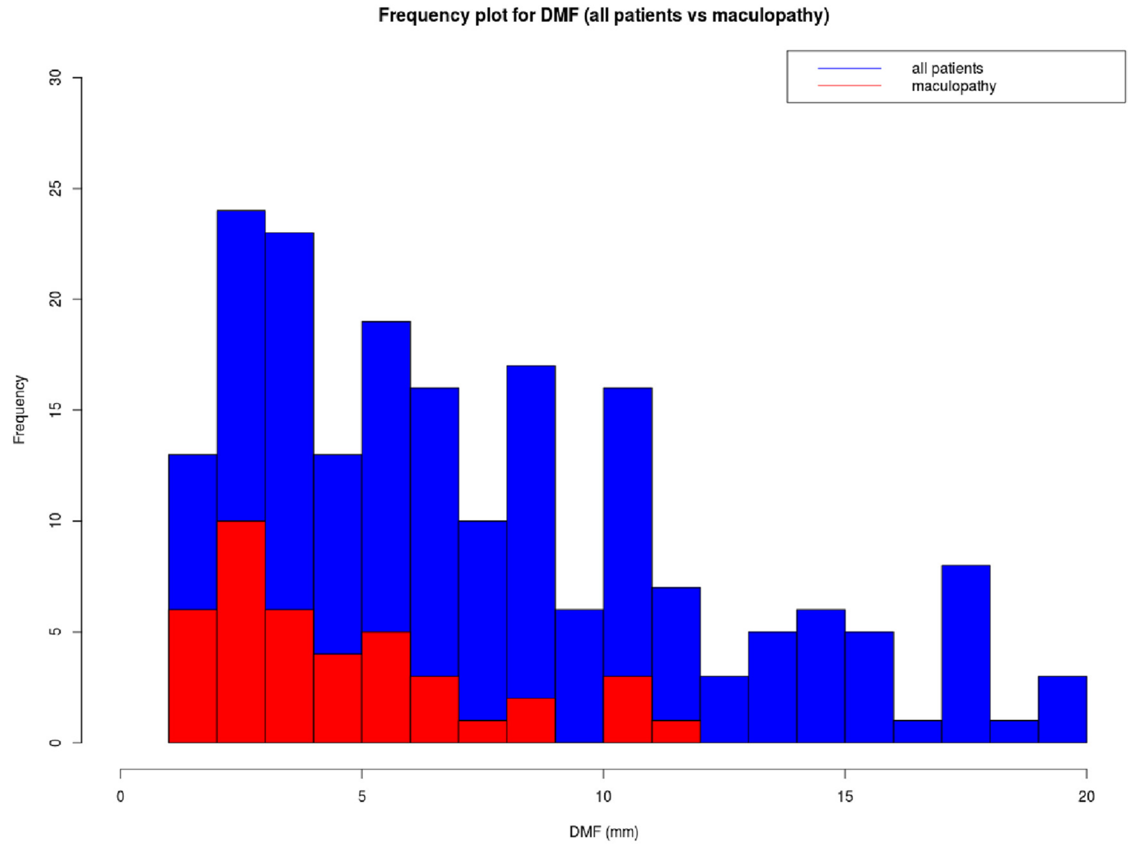

Fig. 2. Graph showing the incidence of radiation maculopathy as a function of distance between the posterior margin of the tumor and the fovea (distance between margin of the tumor and fovea). Y axis: number of patients. X axis: distance between the posterior margin of the tumor and the fovea (distance between margin of the tumor and fovea). Blue represents total number of patients; and red represents number of patients who developed radiation maculopathy. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

There are few successful treatments for patients who develop the radiation maculopathy. Therefore, the most prudent strategy is to reduce or avoid its occurrence, if possible, to identify patients at higher risk of onset.

Our results, as others reported in the literature, suggest that radiation dose to the fovea can help to predict the risk of developing radiation maculopathy as well as the dose to the lens and optic disc for developing cataract and optic neuropathy.

In particular, we identified a threshold dose to the fovea of 50 Gy; below this value, the risk associated to the development of radiation maculopathy was 12%; above it, the associated risk was 46% on average (range 31−60%), and this result was statistically significant.

Other studies had demonstrated that a higher radiation dose increases the risk for radiation maculopathy. Stack *et al.* found 63% risk for radiation maculopathy if the dose to the macula exceeded 90 Gy (15).

Brown *et al.* and Gunduz *et al.* suggested that a cutoff dose of approximately 5000 cGy or less may be tolerated safely by the fovea following plaque radiotherapy (3, 16).

An attempt to reduce macular irradiation, and minimize the potential loss of visual acuity, could be made by reducing the dose delivered to the tumor. A retrospective analysis of radiation dose at the tumor apex was performed by Perez *et al.* in 190 patients with uveal melanoma treated with episcleral I-125 brachytherapy. Doses ranged from less than 65 Gy to greater than 85 Gy. No relationship between dose delivered to the apex and local control and distant metastases was found, whereas a significant relationship between radiation dose and ocular toxicity was identified (17).

Other studies have shown similar results. A randomized controlled trial of varying radiation doses in the treatment of choroidal melanoma by Gragoudas *et al.* found that lower doses of radiation correlated with lower rates of visual field loss and radiation optic neuropathy, without compromising local tumor control or metastatic spread (18).

Murray *et al.* demonstrated that use of less radiation might lower the incidence of radiation-related sight-threatening complications, an equally effective treatment in local tumor control and prevention of metastasis when compared with standard treatment doses (19). Therefore, to minimize the radiation damage, it might be useful to prescribe a lower dose to the tumor apex for lesions close to the macula.

Another way to decrease the dose-related morbidity is to use neoadjuvant treatment to brachytherapy with the aim of reducing tumor volume and hence the dose delivered to the tumor apex, as we demonstrated in amelanotic melanomas using the prebrachytherapy photodynamic therapy treatment (20).

In the literature, possible modifications to the dose to the fovea are reported using different approaches. According to Russo *et al.*, eccentric plaque placement provides acceptable rates of local tumor control and, if the tumor does not extend closer than 3 mm of fovea, relatively good rates of visual conservation (21). New ophthalmic radiation plaque was proposed by Marwaha *et al.* to deliver less radiation exposure to critical vision structures than Collaborative Ocular Melanoma Study treatment plaques while still delivering the same therapeutic dose (22).

Another risk factor for the onset of radiation maculopathy is the distance between the tumor margin and fovea, as already documented by Finger who noted that plaque position affected the incidence and location of radiation-related complications within the eye. Specifically, using the ophthalmic equator as a dividing line, posterior ocular plaques were associated with relatively high incidence of maculopathy (52%) as compared with anterior tumors (4%) (23).

Examining the distance to the fovea, we found that no anterior tumor, with iridic or iridociliary location, developed radiation maculopathy. Besides, we found a cutoff of 11.22 mm of distance between posterior tumor margin and the fovea; beyond this value, no patients developed radiation maculopathy.

In our study, significant factors predictive of the development of radiation maculopathy were tumor height, dose to the fovea, and distance to the fovea, as shown in Table 3. In fact, only 10% of patients receiving a dose to the fovea lower than 50 Gy developed radiation maculopathy, and if the tumor margin was 11.22 mm from the fovea, no maculopathy was detected. The median time to develop maculopathy was found to be 31 months after plaque treatment.

Three years after Ru-106 plaque, 30.9% of our patients developed radiation-related complications (25.5% radiation maculopathy and 5.4% optic neuropathy), while 69.1% of the irradiated eyes maintained a stable visual acuity.

In our patients, the local recurrence rates were 2.2%, 5.2%, and 8.3% at 12, 24, and 36 months after irradiation, respectively. This is well in range of local tumor control rates reported in the literature for tumors with comparable size and treated with the same isotope. The weighted mean local failure rate among seven studies using Ru-106 brachytherapy for tumors with a weighted mean tumor largest basal diameter of 10.9 mm and a weighted mean tumor height of 4.1 mm was reported to be 9.6% (24).

The rate of distant metastatic disease in this work was 3.3% at 3 years; these findings are comparable to similar findings in other published studies (25).

This study presents the experience of a single center using Ru-106 in the treatment of small−medium uveal melanoma. Our findings underline the importance of dosimetric data and location of tumor as main parameters in helping to predict the risk of radiation toxicities, and highlight the need to adopt strategies aimed at reducing the total dose, without compromising the effectiveness of the treatment.

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